

The Herd and the Prion

A bovine oxygen carrier begins in a pasture, and the pasture is where its hardest safety argument is won or lost

Every conversation about a bovine hemoglobin oxygen carrier eventually reaches a question that the listener is usually too polite to ask. It comes from cattle. Cattle carry a disease that frightened a continent. Why should anyone be comfortable with a medicine that starts in a cow?

The polite answer is that the risk is negligible and the product is purified. That answer may well be right, but it is a conclusion and not an argument, and a patient community deserves the argument. This essay walks through the argument the way a supply-chain engineer would, from the pasture to the vial, because the safety of a BHOC is not a single property of the molecule. It is a property of an entire chain of custody, and every link in that chain has its own physics, its own failure modes and its own audit trail.

What the fear is actually about

Bovine spongiform encephalopathy, usually shortened to BSE, was recognized in British cattle in the mid-1980s. It belongs to a family of diseases caused by prions, misfolded forms of a normal host protein that convert their normal neighbors into more of the misfolded form. Prions are not bacteria or viruses. They carry no genome. They resist heat, radiation, formaldehyde and most of the tools that sterilize ordinary pathogens, which is precisely why the disease became a public health crisis rather than an agricultural nuisance.

The turning point came in 1996, when British authorities reported that a new human disease, variant Creutzfeldt-Jakob disease, was linked to exposure to the BSE agent. The cattle epidemic had been amplified by feeding ruminant-derived protein back to ruminants, a practice later banned. The human cases followed dietary exposure, chiefly to neural tissue that entered meat products. The United States identified its first case of BSE in December 2003, in a cow that had been imported from Canada, and regulators on both sides of the Atlantic rebuilt their rules around a single insight: the infectious agent is not spread evenly through the animal.

That insight is the foundation of everything that follows. In an infected animal the agent concentrates in the brain, the spinal cord, the eyes, certain ganglia and the distal portion of the small intestine. It is not concentrated in muscle, and in the bioassay experiments that have been done, infectivity has not been detected in cattle blood. That last statement deserves care. Absence of detected infectivity in an assay is not proof of absence, because every assay has a limit of detection, and prion assays in particular are slow, expensive and species-dependent. It is evidence, not a guarantee, and a careful writer says so.

The awkward human comparison

There is a comparison here that the bovine skeptic seldom hears, and I find it clarifying. The only documented cases of person-to-person transmission of variant Creutzfeldt-Jakob disease through blood came through human blood. In the United Kingdom, a small number of probable

transfusion-transmitted infections were identified, on the order of a handful, in recipients of red cells donated by people who were later diagnosed with the disease. Those donors were incubating the infection silently, and the cells had not been filtered for white cells, which are thought to carry more of the agent than the red cells themselves.

The consequence is a strange asymmetry. Society accepted human blood, which has a demonstrated if rare route for prion transmission, and built screening rules around it: deferral of donors who had lived in the United Kingdom during the peak years, leukoreduction of cell products, and lifetime deferral of anyone who had received a transfusion there. Meanwhile, a bovine product has no demonstrated route for transmission at all, but its source animal carries the name of the disease. The perception of risk and the measured risk point in opposite directions.

None of this means a BHOC is risk-free. It means the risk should be examined where it actually lives, and it lives in three places.

Link one: the animal

The first defense is the choice of animal, and it has three parts.

The first is geography and history. A herd raised in a country or region with a very low or zero record of BSE, with documented feed controls, starts with a low prior probability of infection. When a manufacturer describes a herd as closed, it means the animals were born into it and no outside cattle entered, so that a contaminated animal cannot be introduced.

The second is age. BSE has a long incubation period, typically measured in years, and detectable disease in young animals is vanishingly rare. Restricting collection to animals below a certain age is a blunt but effective tool.

The third is feed. The epidemic was driven by rendered ruminant protein in cattle feed. A herd raised on pasture and plant feed, with an auditable record of what each animal ate, removes the mechanism that amplified the disease in the first place.

What I find striking about these controls is that they are cheap once they are designed into the operation and enormously expensive to add afterward. A company that builds a BHOC around a certified herd from the start has a very different risk profile from one that buys blood on the spot market. Regulators know this, which is why documentation of the source herd, its records and its veterinary oversight sits near the front of any submission.

Link two: the moment of collection

The second link is where an unexpectedly physical hazard hides, and it is the detail I have never seen given the attention it deserves.

To collect blood from a slaughtered animal, the animal must first be stunned. One family of stunning methods uses a captive bolt, a device that drives a rod into the skull. Studies of that method have shown that it can push fragments of brain tissue into the circulation, where they can appear in the blood draining from the neck. Some designs inject compressed air into the cranial cavity, which increases the risk further. Regulators in several jurisdictions restricted or prohibited the more disruptive techniques for this reason.

Think about what that means. In an infected animal, the very tissue where prions concentrate could reach the blood at the moment of collection. The blood itself may carry little or no agent, but the process of obtaining it could add some. That is why a well-designed collection protocol treats stunning as a design variable. It specifies non-penetrating stunning devices, avoids methods that disrupt the brain, and monitors collected blood for markers of central nervous system tissue contamination. This is a subtle point, and it is one that an outsider would not think to ask about, yet it separates a careful supply chain from a casual one.

A related discipline concerns the animal's inspection. Animals that are unwell, that cannot walk unassisted or that show neurologic signs are excluded, because those are the animals in which any residual disease risk concentrates. The rules regulators wrote for food after 2003 embody the same idea, banning material from what they call non-ambulatory disabled cattle and removing high-risk tissues from all older animals.

Link three: the process

The third link is manufacturing. A purified hemoglobin solution is not blood. It is the product of a sequence of steps that begins with red cells and ends with a protein in a defined solution. The red cells are separated from plasma and washed. They are broken open. The membranes and other cellular debris are removed by filtration. The hemoglobin is then purified, often by chromatography, and concentrated. It is chemically polymerized and filtered again, and finally it is packaged.

Every one of those steps removes something. The question for prion safety is how much. The answer is measured in what regulators call log reduction. In a validation study, a manufacturer deliberately spikes a sample of intermediate material with a known amount of a prion preparation, runs it through the step, and measures how much survives. A step that removes a thousand-fold has a three-log reduction. Steps in series add their reductions, provided they act by independent mechanisms.

For prions, the workhorse mechanisms are physical partitioning, such as precipitation, filtration and chromatography, which physically separate the protein aggregates from the product, and chemical inactivation, such as treatment with strong alkali, which is one of the few reagents known to degrade prions meaningfully. Published work on the manufacture of human plasma-derived proteins has found that some individual steps deliver several logs of clearance, which is why plasma product makers rely on those numbers. A BHOC manufacturer borrows the same toolkit and the same validation logic.

I want to be circumspect here. Clearance studies are model experiments. They depend on the spiking material resembling the natural agent, on the scaled-down process resembling the real one, and on assays that are sensitive enough. Regulators accept them because there is no better alternative, and because the evidence base for individual steps has grown over three decades. But a thoughtful advocate should describe them as a layer of risk reduction that adds to the earlier layers, not as a proof.

The other pathogens nobody mentions

Prions dominate the conversation because of their cultural weight, but they are not the only concern in a bovine product, and it would be dishonest to leave the others out.

Cattle harbor viruses. The one most familiar to pharmaceutical manufacturers is bovine viral diarrhea virus, which has long been a known contaminant of bovine-derived reagents such as fetal calf serum used in cell culture. It does not infect people readily, but its presence would signal a failure of upstream controls, and manufacturers test for it. Others include bovine parvovirus and a group of enveloped viruses, and the manufacturing process needs steps demonstrated to remove or inactivate both enveloped and non-enveloped types.

There are also endotoxins, fragments of bacterial cell walls that can trigger fever and hypotension even in tiny amounts. Blood collected from a slaughterhouse floor is exposed to environmental bacteria, and a product intended for intravenous use must meet strict endotoxin limits. Endotoxin is a plausible confounder in the early hemoglobin trials, because contamination in a first-generation product can produce a syndrome that looks like the effects of the molecule itself. The field has spent years learning to separate the two.

Finally there is the problem of the protein itself. Bovine hemoglobin is a foreign protein. It is more similar to human hemoglobin than a worm protein would be, but it is not human. Antibodies against it can develop, and a patient who received it once may react differently the second time. This is an issue of immunology rather than infection, and it belongs in the same conversation because it is part of what it means to be a bovine product.

Why one herd is not enough

Engineers speak of single points of failure, and a BHOC supply chain can have several. The most obvious is a single source herd. A disease outbreak in cattle, whether or not it involves BSE, could halt collection for months. A regulatory change in the source country could make the herd ineligible. A single processing plant, a single supplier of the polymerizing reagent and a single sterile filling line are further examples.

The history of Biopure, the American company that developed Oxyglobin and Hemopure, is instructive. It began in the 1980s by sourcing bovine blood from regional slaughterhouses, and over time it moved toward tightly controlled herd sourcing. Its later story, ending in a bankruptcy filing in 2009 and a sale of assets, was driven by financial and regulatory problems rather than a prion event. Yet even that lesson applies: when a company is thin, its supply chain is thin, because redundancy costs money.

An advocate who wants access to a BHOC should therefore ask not only whether it is safe today, but whether the chain that makes it can survive a shock. Does the manufacturer have more than one qualified herd? Are there stored reserves of intermediate material? Is there a plan for the day an animal in the herd tests positive for anything? A supplier that cannot answer those questions has a resilience problem even if the product is excellent.

Reading a batch record as a detective would

Consider what a single vial of a BHOC should be able to tell you if you interrogated it. The lot number on its label leads to a batch record, and the batch record leads back through the purification steps to the pooled starting material, and the pool leads to the individual collection days, and each collection day leads to a list of animals with ear tags, birth dates, herd of origin and inspection results. In a well-run operation, that chain can be walked in either direction

within hours. If an animal were later found to be infected with anything, the manufacturer could name every lot that contained its blood and every patient who received those lots.

This is the property called traceability, and it is the reason I trust documentation more than reassurance. A system that can be audited backward can also be corrected forward. It is worth noticing that this is exactly the property human blood banks acquired only after painful lessons in the 1980s, and that it was imposed on them from outside. A bovine product designed from the beginning with the same discipline starts ahead of the game.

The reverse detective exercise is equally revealing. Imagine a manufacturer whose records show blood pooled from many slaughterhouses with no animal-level identification. Even if every animal was healthy, the manufacturer could not prove it, and any adverse event would be untraceable. Regulators are right to treat that pattern as a serious deficiency, and patient advocates are right to ask about it. The question to put to any developer is short and plain: if a problem were found in one animal five years from now, how many minutes would it take you to find every patient exposed to its blood?

What the regulators ask

The regulatory frame in the United States is a layered one. The Food and Drug Administration has rules, first written in interim form and later finalized, restricting the use of certain cattle materials in food and cosmetics, and it has guidance on cattle-derived materials in medical products that requires manufacturers to document the country of origin, the age of the animals, the tissues used and the steps that reduce risk. Its position on the food supply is that the risk of human exposure through regulated products is negligible given the controls in place. A medical product goes through a stricter dossier than a food, because it is injected.

I would summarize what a regulator wants from a bovine-derived injectable in five sentences. Show me where the animals came from. Show me what they ate. Show me how they were inspected and how the blood was collected. Show me what your process removes, with validation data. Show me how you would know if something changed. Every element is documentation, and every element can be audited.

What this means for a patient audience

A patient community will hear two competing simplifications. One says cows carry mad cow disease, so bovine products are dangerous. The other says purification makes everything safe, so there is nothing to worry about. Both are wrong in the same way: they collapse a chain into a single link.

The accurate position is more useful. The prion risk of a BHOC is the product of several independent probabilities: that the animal is infected, that infectivity reaches the collected blood, that it survives the process, and that a dose sufficient to infect reaches a recipient by the intravenous route. Each factor is small, and the layered controls are designed so that no single failure is decisive. That does not equal zero. It equals a risk small enough that regulators and manufacturers of other bovine-derived injectables have accepted it for decades, and it is watched by a surveillance system that would notice a signal.

I would encourage three practical requests for any organization advocating access. Ask for the herd documentation, in summary form if the details are proprietary. Ask for the prion and viral clearance data, at least the number of logs achieved by each validated step. And ask for the plan for lot traceability, so that if a problem emerged years later, each vial could be traced back to an animal and a date. A manufacturer that answers those three questions readily is signaling a mature operation. One that deflects them is signaling the opposite.

The strangest lesson from this subject is a reversal of the usual intuition. We tend to trust what feels natural and fear what feels foreign, and human blood feels safer than cow blood. The record says the opposite is at least arguable. Human blood has carried a demonstrated prion route, while a carefully sourced bovine product has none on record. Feelings are a poor instrument for measuring that, and audit trails are a good one. The pasture is where the argument starts, and the paperwork is where it ends.

Further reading

Regulatory background is available from the US Food and Drug Administration on bovine spongiform encephalopathy and on cattle-derived materials in medical products, including the 2007 rule on materials derived from cattle used in medical products and the 2016 final rule on cattle materials in food and cosmetics. Reviews of prion clearance during plasma protein manufacture describe the validation approach. Details of stunning methods and central nervous system emboli appear in the veterinary and food safety literature. Figures for the small number of transfusion-transmitted variant Creutzfeldt-Jakob disease cases come from United Kingdom surveillance reports and should be confirmed there.